

# Hartshorne Chapter 2 Section 3: Properties of Schemes

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The main goal for this section is understanding the topological / algebraic properties on schemes.

## 1 Basic Properties, and their Relations

### Definition: Connected Scheme (Topological)

A scheme is **connected** if its topological space is connected.

### Definition: Irreducible Scheme (Topological)

A scheme is **irreducible** if its topological space is connected.

### Definition: Reduced Scheme (Algebraic)

A scheme is **reduced** if every open subset  $U \subseteq X$  has  $\mathcal{O}_X(U)$  being reduced.

### Definition: Integral Scheme (Algebraic)

A scheme is **integral** if every open subset  $U \subseteq X$  has  $\mathcal{O}_X(U)$  being an integral domain.

As an examples / exercise, here are some properties of affine schemes:

### Proposition

Given  $X = \text{Spec}(A)$  an affine scheme, then:

1.  $X$  is connected iff  $A \not\cong B \times C$  as commutative rings.
2.  $X$  is irreducible iff  $\text{Nil}(A)$  is prime
3.  $X$  is reduced iff  $\text{Nil}(A) = 0$
4.  $X$  is integral iff  $A$  is an integral domain.

### Proof:

1. It's easier to prove when assuming  $A$  is already reduced, and also  $X$  is disconnected iff  $A \cong B \times C$  as commutative rings.

If  $X$  is disconnected, then consider the two radicals  $I, J \subseteq A$  that satisfies  $X = V(I) \sqcup V(J)$ . Since  $V(I) \cap V(J) = V(I + J) = \emptyset$ , one must have  $I + J = A$ ; on the other hand,  $V(I) \sqcup V(J) = V(I \cap J) = X = \text{Spec}(A)$  (while  $I, J$  are radicals, implying  $I \cap J$  is radical)

shows that  $I \cap J = \text{Nil}(A) = 0$  (by assumption). So, one has  $A \cong I \oplus J$  as  $A$ -module, in particular each  $I, J$  is itself a ring (since  $J \cong A/I$ ).

Then, for non-reduced case, if  $A \twoheadrightarrow A/\text{Nil}(A) \cong \overline{B} \times \overline{C}$ , let  $B, C$  be the preimage of  $\overline{B}, \overline{C}$  respectively, we ought to show  $B \times C \twoheadrightarrow A$  (then since the ideal in  $B \times C$  are of the form  $I_B \times I_C$ , one has  $A \cong B/I_B \times C/I_C$ , where  $I_B \times I_C$  is the kernel). However, it's simple by  $(b, c) \mapsto b + c$

Suppose  $A \cong B \times C$ , then  $B \hookrightarrow B \times C$  and  $C \hookrightarrow B \times C$  are ideals, satisfying  $B + C = A$  and  $B \cap C = 0$ , showing  $V(B) \cap V(C) = V(B + C) = V(A) = \emptyset$ , and  $V(B) \sqcup V(C) = V(B \cap C) = V(0) = \text{Spec}(A)$ . So,  $\text{Spec}(A)$  is disconnected.

2. Since one has  $\text{Spec}(A) = V(\text{Nil}(A))$ , then one has  $\text{Spec}(A) \cong \text{Spec}(A/\text{Nil}(A))$  as topological space. Which, if  $\text{Nil}(A)$  is prime,  $A/\text{Nil}(A)$  is an integral domain, hence  $\text{Spec}(A)$  is irreducible; conversely, suppose  $\text{Spec}(A) \cong \text{Spec}(A/\text{Nil}(A))$  is irreducible, but  $\text{Nil}(A)$  not integral domain, there exists  $i, j \in A \setminus \text{Nil}(A)$ , such that  $\bar{i} \cdot \bar{j} = 0$ . As a result, one has  $\text{Spec}(A/\text{Nil}(A)) = V((0)) = V((\bar{i})(\bar{j})) = V((\bar{i})) \cup V((\bar{j}))$ , while each  $V((\bar{i})), V((\bar{j}))$  are proper, contradicting the irreducibility. Hence,  $\text{Nil}(A)$  must be prime.

3. For reduced property, we'll verify that they're local properties first:

- First, recall that  $\text{Nil}(A_P) = (\text{Nil}(A))_P$  (by the fact that extension of ideals under localization preserves intersection, and any other prime ideals not containing  $P$  extends to unit ideal).

Hence, one has  $\text{Nil}(A) = 0$  (as  $A$ -modules)  $\iff \text{Nil}(A_P) = \text{Nil}(A)_P = 0$  (as  $A_P$ -module) for all prime ideal  $P \subseteq A$  (this is the module property).

Now, since every open subset  $U \subseteq X$  has  $\mathcal{O}_X(U) = \{s : U \rightarrow \bigsqcup_{P \in U} A_P\}$ , since  $A$  is reduced  $\iff A_P$  being reduced, it enforces  $\mathcal{O}_X(U)$  to be reduced also (since to be nilpotent it's coordinate-wise nilpotent, which is 0).

Conversely, suppose  $A$  is not reduced, then one of  $A_P$  has some  $a \in A$  such that  $\frac{a}{1} \neq 0$  but  $(\frac{a}{1})^n = 0$  in  $A_P$ , or  $a^n u = 0$  for some  $u \in A \setminus P$ . Then, take the open subset  $D(u)$ , one has the function  $s : D(u) \rightarrow \bigsqcup_{Q \in D(u)} A_Q$  by  $s(Q) = \frac{a}{1}$  satisfies  $(\frac{a}{1})^n = \frac{a^n}{1} = 0$  (since  $a^n u = 0$ , with  $u \notin Q$  for all  $Q \in D(u)$ ). Then,  $s \neq 0$  (since  $s(P) = \frac{a}{1} \neq 0 \in A_P$ ), while  $s^n = 0$ . So,  $\mathcal{O}_X(D(u))$  is not reduced.

4. If  $X$  is integral, it's obvious that  $\mathcal{O}_X(X) = A$  is integral domain.

The converse will be proven in the next proposition (for general scheme) (where a scheme is integral  $\iff$  it's reduced and irreducible. Here, reduced and irreducibility both come from the assumption that  $A$  is integral domain).

**(Remark:** The general case for non-reduced version of 1 is not solved.)  $\square$

Now, here's the general statement on the scheme:

### *Proposition*

A scheme  $X$  is integral iff it is both reduced and irreducible.

**Proof:**  $\implies$ :

Suppose  $X$  is integral, it's clear that it's reduced (as all integral domain is reduced).

For irreducibility, suppose the contrary that it's reducible. Hence, there exists two nonempty open subsets  $U, V \subseteq X$ , where  $U \cap V = \emptyset$ . As a result, take any  $s_u \in \mathcal{O}_X(U)$  and  $s_v \in \mathcal{O}_X(V)$ , since their domain has empty intersection, they can always be glued together. So, take  $(s_u, 0_v)$  as the gluing of nonzero  $s_u$  with  $0_v \in \mathcal{O}_X(V)$ ,  $(0_u, s_v)$  as the gluing of nonzero  $s_v$  with  $0_u \in \mathcal{O}_X(U)$ . The

product of the two is 0 (as  $0_u$  evaluates to 0 on  $U$ , and similarly for  $0_v$ ), while each individual of them is not, reaching a contradiction that  $\mathcal{O}_X(U \sqcup V)$  is an integral domain.

(Even stronger, one can say  $\mathcal{O}_X(U \sqcup V) \cong \mathcal{O}_X(U) \times \mathcal{O}_X(V)$  if  $U \cap V = \emptyset$ , as the two functions share no domain, so has no affect on each other).

$\Leftarrow$ :

Conversely, suppose  $X$  is reduced and irreducible. Let  $U \subseteq X$  be any open subset, and suppose  $f, g \in \mathcal{O}(U)$  satisfies  $fg = 0$ . Define the set  $Z_f := \{x \in U \mid f_x \in m_x\}$  (where  $f_x \in \mathcal{O}_{X,x}$  and  $m_x$  is the maximal ideal of  $\mathcal{O}_{X,x}$ ), and similarly  $Z_g := \{x \in U \mid g_x \in m_x\}$ . Note that since  $f_x g_x = 0 \in m_x$  for all  $x \in U$ , each  $x \in U$  has  $f_x \in m_x$  or  $g_x \in m_x$ , showing  $Z_f \cup Z_g = U$ .

On the other hand, they're actually closed subsets: By intersecting with affine covers, say  $U$  is affine (then  $U = \text{Spec}(\mathcal{O}_X(U))$ ), then the complement  $X_f := U \setminus Z_f$  are all prime ideals  $P \in U$ , such that  $f_P$  is a unit (in particular, one has  $f \notin P$ , as  $\mathcal{O}_{X,P}$  is the localization of  $\mathcal{O}_X(U)$  at  $P$ , then  $f_P$  is a unit  $\implies f$  can't be contained in  $P$ ). So, working it out set wise, we have  $X_f = D(f)$  in fact, which is open in  $U$ . So,  $Z_f$  is closed in  $U$ . And, since this is true for arbitrary affine cover,  $Z_f$  must be closed in  $X$  itself.

So, we have  $Z_f, Z_g$  two closed subset such that  $Z_f \cup Z_g = U$ . With  $X$  irreducible, any of its open subset must also be irreducible, showing  $Z_f = U$  or  $Z_g = U$ .

Suppose  $Z_f = U$ , then this implies  $X_f = D(f) = \emptyset$  when restricting it to  $U$ 's intersection with any affine cover, as a result  $f$  must be nilpotent in all of these affine covers, hence nilpotent in  $U$ . But, using the reduced assumption, this implies  $f = 0$ . Which, it proves the integrality.  $\square$

Intuition wise, we can see it relies heavily on affine schemes. If possible, in the future we'll practice more on how restricting to affine scheme works.

## 2 Noetherian Property

Similar to how in classical sense we worked with  $k[x_1, \dots, x_n]$  (a Noetherian ring), we wish to work with some similar ideas in scheme theory also.

### Definition: Locally Noetherian, Noetherian Scheme

A scheme  $X$  is **locally Noetherian**, if there exists an open cover  $\{U_i\}$  of  $X$ , such that each  $U_i = \text{Spec}(A_i)$  (affine), and each  $A_i$  is a Noetherian ring.

Similarly,  $X$  is **Noetherian**, if it's locally Noetherian and quasi-compact. As an equivalent definition,  $X$  is Noetherian, if it can be covered by finitely many affine open subsets, with each affine subset corresponds to a Noetherian ring.

### Remark

If  $X$  is a Noetherian scheme, then it's clear it's Noetherian as topological space (since it's finite union of  $\text{Spec}(A_i)$ , which  $A_i$  is Noetherian ring guarantees  $\text{Spec}(A_i)$  is a Noetherian topological space; so, finite union is still Noetherian topological space).

Yet, the converse is false:

Consider the ring  $A = k[x_1, x_2, \dots]/(x_1^2, x_2^2, \dots)$ . Notice that all  $\overline{x_i} \in A$  are nilpotent, hence  $(\overline{x_1}, \overline{x_2}, \dots) \subseteq \text{Nil}(A)$ ; on the other hand, it's also a maximal ideal (as the quotient  $k[x_1, x_2, \dots] \twoheadrightarrow k$  by sending all  $x_i \mapsto 0$  contains the ideal  $(x_1^2, x_2^2, \dots)$ , hence factors through  $A$ ; however, in here any constant can't be sent to 0, while all  $\overline{x_i} \mapsto 0$ , so the kernel through  $A$  must be  $(\overline{x_1}, \overline{x_2}, \dots)$ , showing that  $A/(\overline{x_1}, \overline{x_2}, \dots) \cong k$ , hence it's also maximal). As a result,  $\text{Spec}(A) = \{*\}$  (since it must only be the maximal ideal), which is trivially Noetherian.

On the other hand, the ring  $A$  can never be Noetherian, since the chain  $(\overline{x_1}) \subset (\overline{x_1}, \overline{x_2}) \subset \dots$  pulls back to  $(x_1) \subset (x_1, x_2) \subset \dots$  in  $k[x_1, x_2, \dots]$  (which is a strict chain), so it must also be a strict chain in  $A$ , showing it's not Noetherian as ring.

Finally,  $\text{Spec}(A)$  is not Noetherian as schemes, since the only open affine subset is itself, and the ring is  $A$  (which is not Noetherian).

This definition is good in an algebraic sense, as one can always find an affine neighborhood associated to a Noetherian ring. Yet, it'll be a lot more convenient if one can choose any affine open neighborhood to be associated to Noetherian rings. This is indeed the case in general:

### Proposition

A scheme  $X$  is locally Noetherian iff every open affine subset  $U \subseteq X$ ,  $U \cong \text{Spec}(A)$  has  $A$  being a Noetherian ring.

In particular, if  $X = \text{Spec}(A)$ , then it's a Noetherian scheme iff  $A$  is Noetherian.

### Proof: $\Leftarrow$ :

Suppose every open affine subset  $U \cong \text{Spec}(A)$  for some Noetherian ring  $A$ , it's clear  $X$  is a locally Noetherian scheme by definition.

$\Rightarrow$ :

Suppose  $X$  is locally Noetherian, and given any open affine subset  $U \cong \text{Spec}(A)$  (i.e. the restriction of schemes has  $(U, \mathcal{O}_U) \cong (\text{Spec}(A), \mathcal{O}_A)$  as locally ringed space). Choose  $\{U_i\}$  as the affine cover of  $X$  so that each  $U_i \cong \text{Spec}(B_i)$  for some Noetherian  $B_i$ , and consider the intersection  $U \cap U_i \subseteq \text{Spec}(B_i)$ . Since within  $\text{Spec}(B_i)$  this is open, by quasi-compactness of prime spectrum, there exists finitely many  $f_{i,1}, \dots, f_{i,j_i} \in B_i$ , so that  $U \cap U_i = D(f_{i,1}) \cup \dots \cup D(f_{i,j_i})$ , and each  $D(f_{i,l})$  has  $\mathcal{O}_A(D(f_{i,l})) \cong (B_i)_{f_{i,l}} \cong B_i[x]/(xf_{i,l} - 1)$ , which by the Noetherianness of  $B_i$ , this ring is also Noetherian.

As a result, we swap  $U \cong \text{Spec}(A)$  with all of the following:

$$U = \bigcup_i U \cap U_i = \bigcup_i \left( \bigcup_{l=1}^{j_i} D(f_{i,l}) \right) \quad (2.1)$$

Where, each  $D(f_{i,l})$  corresponds to a Noetherian ring (and by the isomorphism of schemes, one has  $\mathcal{O}_X(D(f_{i,l})) \cong (B_i)_{f_{i,l}}$ ). Hence, the statement also reduces to: If  $X = \text{Spec}(A)$  is an affine scheme covered by affine opens that're Noetherian rings' spectra, then  $A$  is Noetherian.

Suppose  $U \subseteq \text{Spec}(A) = X$  is an affine open subset where  $U \cong \text{Spec}(B)$  (with  $B$  Noetherian ring), this inclusion  $U \hookrightarrow X$  induces a ring homomorphism  $A \rightarrow B$ . Now, by openness of  $U$ , by the fundamental open sets, let  $f \in A$  satisfies  $D(f) \subseteq U$ ; let  $\bar{f} \in B$  be the image of  $f$  under the ring homomorphism, then the fundamental open's property shows that  $\iota^{-1}(D(f)) = D(\bar{f})$ , in particular one has  $\mathcal{O}_B(D(\bar{f})) \cong \mathcal{O}_A(D(f))$ , so  $B_{\bar{f}} \cong A_f$ , showing  $A_f$  is Noetherian.

This again swaps  $X$  with a collection of  $f \in A$ , such that  $D(f)$  associates to  $A_f$  a Noetherian ring. Using the quasi-compactness of  $X = \text{Spec}(A)$ , there exists  $f_1, \dots, f_n \in A$ , such that  $\bigcup_{i=1}^n D(f_i) = X$ , and each  $A_{f_i}$  is Noetherian. As a result,  $(f_1, \dots, f_n) = A$ . Now, let's do a lemma:

### Lemma

Let  $I \subseteq A$  be an ideal, and  $\varphi_i : \rightarrow A_{f_i}$  be the localization map for each  $i \in \{1, \dots, n\}$ . Then:

$$I = \bigcap_{i=1}^n I^{ec} = \bigcap_{i=1}^n \varphi_i^{-1}(\varphi_i(I)A_{f_i}) \quad (2.2)$$

**Proof:** Let's recall that each  $\varphi_i^{-1}(\varphi_i(I)A_{f_i}) = \{r \in A \mid \exists k \in \mathbb{N}, r f_i^k \in I\}$  (since the extension then contraction is all the element, such that multiplied by some element in the multiplicative set it lies in  $I$ ). So, this shows the inclusion  $I \subseteq \bigcap_{i=1}^n \varphi_i^{-1}(\varphi_i(I)A_{f_i})$ .

Now, suppose  $r \in \bigcap_{i=1}^n \varphi_i^{-1}(\varphi_i(I)A_{f_i})$ , then there exists  $k_1, \dots, k_n \in \mathbb{N}$  and  $a_1, \dots, a_n \in I$ , such that  $\varphi_i(r) = \frac{a_i}{f_i^{k_i}} \in A_{f_i}$ . In particular, by multiplying suitable  $f_i^{k_i}$  to both the numerator and denominator, WLOG one can assume all  $k_i = k$  for some fixed  $k$ . So, we have  $\varphi_i(r) = \frac{r}{f_i^k} = \frac{a_i}{f_i^k}$ , showing there exists  $m_1, \dots, m_n \in \mathbb{N}$ , such that  $(f_i^k r - a_i) f_i^{m_i} = 0$ . Now, by multiplying new  $f_i$ 's wouldn't affect the equality, hence again WLOG each  $m_i = m$  for some fixed  $m$ .

Herefore, we get  $(f_i^k r - a_i) f_i^m = 0$  for all index  $i$ , showing  $f_i^{k+m} r \in I$ . Now, since  $(f_1, \dots, f_n) \in A$ , notice that  $(f_1^{k+m}, \dots, f_n^{k+m}) = A$  (since its radical contains  $f_1, \dots, f_n$ , hence equals to  $A$ ). Then, for suitable  $b_1, \dots, b_n \in A$ , one has the following:

$$\sum_{i=1}^n b_i f_i^{k+m} = 1, \quad \sum_{i=1}^n b_i f_i^{k+m} r = r \in I \quad (2.3)$$

This finishes the proof.  $\square$

As a result, for any increasing chain of ideals  $I_1 \subset I_2 \subset \dots$ , its image  $\varphi_i(I_1)A_{f_i} \subset \varphi_i(I_2)A_{f_i} \subset \dots$  in  $A_{f_i}$  stabilizes (by a.c.c. on  $A_{f_i}$ ), hence  $\varphi_i^{-1}(\varphi_i(I_1)A_{f_i}) \subset \varphi_i^{-1}(\varphi_i(I_2)A_{f_i}) \subset \dots$  stabilizes in  $A$ .

Then, the lemma states that each  $I_i$  is the intersection of the  $i$ th ideal of a stabilizing chain, hence for large enough  $N$  the  $I_i$ 's chain stabilizes. This shows that  $A$  is Noetherian.  $\square$

For the special case where  $X = \text{Spec}(A)$  is a Noetherian Scheme, locally Noetherianness enforces all rings of affine open subsets to be Noetherian, in particular  $\mathcal{O}_X(X) = A$  is Noetherian.

This demonstrates another important mindset in Algebraic Geometry: Sometimes we're looking at information beyond the space, so being able to transfer between algebraic properties and geometric properties will be crucial.

### 3 Morphisms

Now, we'll start talking about some basic morphisms. Recall that for classical sense, the finite type / finite ring homomorphisms can let us reduce the questions to some quotient of polynomial rings / realizing it as finitely-generated module over the base ring. Here it's similar in scheme theory:

#### Definition: Locally Finite Type / Finite Type Morphism

Given a morphism of schemes  $f : X \rightarrow Y$ , it's called **locally of finite type** if there exists an affine cover  $\{V_i\}$  of  $Y$  (denote each  $V_i = \text{Spec}(A_i)$ ), such that each  $f^{-1}(V_i)$  has an affine cover  $\{U_{ij}\}$  (denote each  $U_{ij} = \text{Spec}(A_{ij})$ ), which the restricted morphism  $f : U_{ij} \rightarrow V_i$  induces a ring homomorphism  $f^\# : A_i \rightarrow A_{ij}$  that realizes  $A_{ij}$  as a finitely-generated  $A_i$ -algebra.

Similarly,  $f$  is called **finite type** if the covering  $f^{-1}(V_i) = \bigcup_j U_{ij}$  can be chosen using finite number of  $j$ .

#### Definition: Finite Morphism

Given a morphism of schemes  $f : X \rightarrow Y$ , it's called **finite** if there exists a affine cover  $\{V_i\}$  of  $Y$  (denote each  $V_i = \text{Spec}(A_i)$ ), such that each  $f^{-1}(V_i)$  is affine (denote  $F^{-1}(V_i) = \text{Spec}(B_i)$ ), where the restricted morphism  $f : f^{-1}(V_i) \rightarrow V_i$  induces a ring homomorphism  $f^\# : A_i \rightarrow B_i$ , realizing  $B_i$  as a finitely-generated  $A_i$ -module.

Again, similar to the previous definition of Locally Noetherian, it in fact enforces all affine open subsets to have the same properties (the proofs are highly nontrivial though...)

Now, let's dive into some properties on open / closed subschemes:

#### Definition: Open Subscheme, Open Immersion

Let  $X$  be a scheme, a scheme  $U$  is an **open subscheme** of  $X$ , if there exists an open subset  $U' \subseteq X$ , such that as schemes one has  $(U, \mathcal{O}_U) \cong (U', \mathcal{O}_X|_{U'})$ .

An **open immersion** is a morphism  $f : X \rightarrow Y$ , such that it can be recognized as the inclusion  $\iota : U \hookrightarrow X$  of open subscheme  $U$  (or, an isomorphism of open subschemes).

#### Definition: Closed Immersion, Closed Subscheme

A morphism of schemes  $f : X \rightarrow Y$  is a **closed immersion** if as topological space,  $X$  is homeomorphic to  $f(X)$  that is a closed subset of  $Y$ , while the induced sheaf morphism  $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$  is surjective.

A **closed subscheme** of  $Y$  is a scheme  $X$  together with a closed immersion  $f : X \rightarrow Y$ . This is also classified up to isomorphism.

As of examples, we have the following:

#### Example: Ring Projection

Given any projection of rings  $A \twoheadrightarrow A/I$ , it induces a homeomorphism  $f : \text{Spec}(A/I) \xrightarrow{\sim} V(I) \hookrightarrow \text{Spec}(A)$ , which is a closed map in topological sense; the reason why it's surjective on the morphism of sheaves, is because for any prime ideal  $P \subseteq A/I$ , with prime ideal  $P^c \subseteq A$ , one has the following commutative diagram:

$$\begin{array}{ccc} A & \twoheadrightarrow & A/I \\ \downarrow & & \downarrow \\ A_{P^c} & \twoheadrightarrow & (A/I)_P \cong A_{P^c}/I_{P^c} \end{array}$$

Where, the property that localization and quotient commutes (as long as the ideal doesn't touch the multiplicatively closed set) shows the bottom map is surjective.

Hence, the morphism  $\mathcal{O}_A \rightarrow f_*\mathcal{O}_{A/I}$  is surjective as sheaves (since it's surjective on stalks, while stalks preserve exactness).

We'd also like to talk about how to „induce“ a scheme structure on a closed subset of a scheme. For open subset it's clear that the choice must be the restriction to subspace topology + the restriction of sheaf, but since there's no notion of sheaf over closed subsets, we do need a bit more work.

#### Example: Reduced Induced Closed Subscheme of Affine Scheme

Given  $X = \text{Spec}(A)$  an affine scheme, and  $Y \subseteq X$  be a closed subset. Then,  $Y = V(I)$  for some ideal  $I$ , which one can take  $Y = V(\sqrt{I})$  (the largest ideal that doesn't change  $Y$ ). Which, using the homeomorphism  $V(\sqrt{I}) \cong A/\sqrt{I}$ , this induces a reduced scheme structure on  $Y$  using  $\sqrt{I}$ .

For general schemes, we require the gluing of schemes as description.

#### Example: Structure on General Schemes

Let  $X$  be arbitrary scheme, and  $Y \subseteq X$  a closed subset. Let  $\{U_i\}$  be an affine cover of  $X$  (with  $U_i = \text{Spec}(A_i)$ ), and consider  $Y_i = Y \cap U_i$  (which is closed in  $U_i$ ), which has an induced reduced structure given for the affines (this structure may be dependent on  $U_i$ ).

Now, to glue the schemes together, we need to argue that any  $Y_i \cap Y_j \subseteq Y_i$  is isomorphic to  $Y_i \cap Y_j \subseteq Y_j$  (since they may induce different structures). Since  $Y_i \cap Y_j = Y \cap (U_i \cap U_j) = Y_i \cap U_j$ , one needs to prove the induced structure is independent of the order of restriction (i.e. one can restrict to one then the other, or vice versa). This suffices to prove the case for the fundamental open subsets.

The full statement is: Given  $U = \text{Spec}(A)$ , if  $f \in A$  and  $V = D(f) = \text{Spec}(A_f)$ , then the reduced induced structure on  $Y \cap U$  restricts to  $Y \cap V$  agrees with  $Y \cap V$ 's reduced induced structure. However, given any ideal  $I \subseteq A$  that's a radical, one has  $I = \bigcap_{P \in V(I)} P$  in  $A$ , when extending it becomes  $I^e = \bigcap_{P \in V(I)} P^e$  in  $A_f$ , which  $P^e$  is prime  $\iff$  its intersection with the set of  $f^n$ 's is empty, or  $P \in D(f)$ . So, going from  $Y \cap U$  (as  $V(I)$ ) to  $Y \cap V$  (as  $V(I^e)$ ) under extension, is the same as the structure on  $Y \cap V$  (as  $V(I) \cap D(f)$ ).

Finally, the gluing condition of sheaves ensure the existence of a structure after gluing all affines together.

This unfortunately is not mentioned in class, which is a bit annoying (since it's still important in our sense).

## 4 Fibre Product

This will be the most complicated part in this section, as the construction is highly nontrivial. The goal is to get the categorical fibre product in the category of schemes. Similar to what we've been doing, it's the matter of reducing down to the fibre products between affine schemes. For this, let's focus on one tool first:

### Theorem: Affinization

Given the functor  $\text{Spec}(\_) : \mathbf{CRing}^{\text{op}} \rightarrow \mathbf{Sch}$ , it has a left adjoint, named „Affinization functor“  $\text{Aff}(\_) : \mathbf{Sch} \rightarrow \mathbf{CRing}^{\text{op}}$ , that's based on the universal property:

Given any scheme  $X$ , its affinization  $\text{Aff}_X$  is an affine scheme with a morphism of schemes  $f : X \rightarrow \text{Aff}_X$ , such that for any affine scheme  $Y$  and morphism of schemes  $g : X \rightarrow Y$ , there exists a unique morphism of affine schemes  $\bar{g} : \text{Aff}_X \rightarrow Y$ , such that  $g = \bar{g} \circ f$ .

$$\begin{array}{ccc} X & \xrightarrow{f} & \text{Aff}_X \\ & \searrow \forall g & \swarrow \exists! \bar{g} \\ & Y & \end{array}$$

This provides a natural isomorphism as follow (where  $\text{Aff}_X = \text{Spec}(B)$  and  $Y = \text{Spec}(A)$ ):

$$\text{Hom}_{\mathbf{CRing}^{\text{op}}}(B, A) \cong \text{Hom}_{\mathbf{AffSch}}(\text{Aff}_X, Y) \cong \text{Hom}_{\mathbf{Sch}}(X, Y) \quad (4.1)$$

**Proof:** For any scheme  $X$ , define the ring  $A_X := \mathcal{O}_X(X)$  (i.e. the ring associated to the global section,  $A_X$  is often denoted as  $\Gamma(X, \mathcal{O}_X)$ , called the „global section functor“). And, define  $\text{Aff}_X := \text{Sp}(A_X)$ .

Which, given any affine scheme  $Y = \text{Spec}(B)$ , and morphism of schemes  $f : X \rightarrow Y$ , it associates with the morphism of sheaves  $f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$ , which  $h = f_Y^\# : \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(f^{-1}(Y)) = \mathcal{O}_X(X)$  is a ring homomorphism  $h : B \rightarrow A_X$ , hence reverses to a unique morphism of schemes on the spectrum level  $h^* : \text{Spec}(A_X) = \text{Aff}_X \rightarrow \text{Spec}(B) = Y$ . So, if such map exists, it must be unique.

Now, we need to show: Given any ring homomorphism  $\alpha : B = \mathcal{O}_Y(Y) \rightarrow A_X = \mathcal{O}_X(X)$ , it defines a morphism of schemes  $\alpha^* : X \rightarrow Y$ . For this, we need gluing of schemes again.

Suppose  $\{U_i\}$  collects all affine open subsets of  $X$ , each  $U_i = \text{Spec}(A_i)$ , then the inclusion  $\iota_i : U_i \hookrightarrow X$  together with  $f : X \rightarrow Y$  composes to  $f \circ \iota_i : U_i \rightarrow Y$ , a morphism between affine schemes, hence it reverses to a ring homomorphism  $\alpha_i^* : B = \mathcal{O}_Y(Y) \rightarrow A_i = \mathcal{O}_X(U_i)$ . Also, the way we define the morphism enforces this to be coming from the composition of  $f_Y^\# : B = \mathcal{O}_Y(Y) \rightarrow A = \mathcal{O}_X(X)$  and  $\iota_i^* : A = \mathcal{O}_X(X) \rightarrow A_i = \mathcal{O}_X(U_i)$ :

$$\begin{array}{ccc} U_i & \xrightarrow{\iota_i} & X \\ & \searrow f \circ \iota_i & \swarrow f \\ & Y & \end{array} \qquad \begin{array}{ccc} A_i & \xleftarrow{\iota_i^*} & A \\ & \swarrow \alpha_i^* & \searrow f_Y^\# \\ & B & \end{array}$$

Now, consider  $U_i \cap U_j$ . As a remark, it may not be affine, but it has affine neighborhoods in each  $U_i, U_j$ , so WLOG assume we're working with the affine neighborhood  $W \subseteq U_i \cap U_j$ . Which, we claim that the structure on  $W$  is independent of  $U_i, U_j$ , and this is because for the two inclusions  $W \hookrightarrow U_i \hookrightarrow X$  and  $W \hookrightarrow U_j \hookrightarrow X$ , the restriction of sheaves forces  $\mathcal{O}_W(W) = \mathcal{O}_X(W)$ , independent of which  $U_i$  it passes through.

In particular, this forces the ring homomorphism  $\alpha_W^* : A = \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_W(W)$  to be compatible in the following sense (where the top is morphism of schemes, the bottom is ring homomorphisms):

$$\begin{array}{ccccc}
& & U_i & & \\
& \nearrow \iota_{W,i} & \downarrow \iota_i & \searrow f \circ \iota_i & \\
W & \xrightarrow{\iota_W} & X & \xrightarrow{f} & Y \\
& \searrow \iota_{W,j} & \uparrow \iota_j & \nearrow f \circ \iota_j & \\
& & U_j & & 
\end{array}$$

$$\begin{array}{ccccc}
& & A_i & & \\
& \nearrow \iota_{W,i}^* & \downarrow \iota_i^* & \nwarrow \alpha_i^* & \\
\mathcal{O}_W(W) & \xleftarrow{\iota_W^*} & A & \xleftarrow{f_Y^\#} & B \\
& \searrow \iota_{W,j}^* & \downarrow \iota_j^* & \swarrow \alpha_j^* & \\
& & A_j & & 
\end{array}$$

So, it's a really large commutative diagram chase between **CRing** and **Sch**.

Which, the morphism  $W \rightarrow Y$  induces a clear ring homomorphism  $B \rightarrow \mathcal{O}_W(W)$  that's compatible with the restriction (and intersection), showing the morphisms could glue together (in a unique way). Globally, this is the morphism  $\alpha : X \rightarrow Y$  (and the property forces it to be unique).

□

With this, we can define the fibre product over affine schemes:

**Lemma: Fibre Product of Affine Schemes**

Given three affine schemes  $X, Y, Z$  (corresponds to ring  $C, B, A$  respectively), and morphisms of schemes  $f : X \rightarrow Z, g : Y \rightarrow Z$  (corresponds to ring homomorphisms  $f^* : A \rightarrow C, g^* : A \rightarrow B$  respectively), then the fibre product in **Sch** exists.

**Proof:** Note that we seek for something stronger: Instead of fibre product over **AffSch**, it's fibre product over **Sch** for any morphisms between affine schemes (so, the universal property must work for any schemes, not just affine schemes). Nonetheless, we can start with just affine schemes:

Since we have the ring homomorphisms going out from  $A$ , it's suitable to construct the fibre coproduct in **CRing**, namely the tensor product:

$$\begin{array}{ccc}
A & \xrightarrow{f^*} & C \\
g^* \downarrow & & \downarrow (g^*)' \\
B & \xrightarrow{\quad} & B \otimes_A C \\
& \searrow (f^*)' & \nearrow \exists! h^* \\
& & D
\end{array}$$

$d_b^*$   $d_c^*$

Which, we define  $X \times_Z Y := \text{Spec}(B \otimes_A C)$ , and  $f' : X \times_Z Y \rightarrow Y, g' : X \times_Z Y \rightarrow X$  be induced by  $(f^*)', (g^*)'$  respectively. Then, the commutativity of the ring homomorphism guarantees the following diagram:

$$\begin{array}{ccc}
Z & \xleftarrow{f} & X \\
\uparrow g & & \uparrow g' \\
Y & \xleftarrow{f'} & X \times_Z Y
\end{array}$$

And, over  $\mathbf{AffSch}$ , it satisfies the universal property, since for any affine schemes  $W$  (say corresponds to ring  $D$ ), together with morphisms of schemes  $d_B : W \rightarrow Y$  and  $d_C : W \rightarrow X$ , such that  $f \circ d_C = g \circ d_B$ . Then, let  $d_B^* : B \rightarrow D$  and  $d_C^* : C \rightarrow D$  denotes the reversed ring homomorphism of the spaces, then one has  $d_C^* \circ f^* = d_B^* \circ g^*$ . Hence, going back to the top diagram, one has a unique ring homomorphism  $h^* : B \otimes_A C \rightarrow D$  by the fibre coproduct property in  $\mathbf{CRing}$ , hence generates a unique morphism of schemes  $h : W = \text{Spec}(D) \rightarrow X \times_Z Y = \text{Spec}(B \otimes_A C)$  (and any such morphism of schemes  $h$  must corresponds to  $h^*$ , showing the uniqueness):

$$\begin{array}{ccc}
Z & \xleftarrow{f} & X \\
\uparrow g & & \uparrow g' \\
Y & \xleftarrow{f'} & X \times_Z Y
\end{array}
\quad
\begin{array}{c}
\curvearrowright d_C \\
\text{---} \exists! h \text{---} \\
\text{---} d_B \text{---} \\
W
\end{array}$$

Basically, it's the reversed diagram of the very first one.

Finally, we ought to show its fibre product property in  $\mathbf{Sch}$ . This can be done through affinization: Suppose  $W$  is arbitrary schemes, with  $d_B : W \rightarrow Y$  and  $d_C : W \rightarrow X$  satisfies  $g \circ d_B = f \circ d_C$ . The universal property of affinization guarantees a factorization  $d'_B : \text{Aff}_W \rightarrow Y$  and  $d'_C : \text{Aff}_W \rightarrow X$  as follow:

$$\begin{array}{ccc}
Z & \xleftarrow{f} & X \\
\uparrow g & & \uparrow g' \\
Y & \xleftarrow{f'} & X \times_Z Y
\end{array}
\quad
\begin{array}{c}
\curvearrowright d_C \\
\text{---} \exists! h \text{---} \\
\text{---} d'_B \text{---} \\
\text{---} d_B \text{---} \\
W
\end{array}$$

Where, the universality of  $X \times_Z Y$  over affine schemes guarantees a unique morphism of schemes  $h : \text{Aff}_W \rightarrow X \times_Z Y$  for the diagram to commute, and composing with the affinization morphism  $a_W : W \rightarrow \text{Aff}_W$ , it generates a morphism for  $W$  to satisfy the fibre product property. It is unique, simply because any morphism of schemes  $W \rightarrow X \times_Z Y$  (where  $X \times_Z Y$  is affine!) must also factor through  $\text{Aff}_W$  uniquely, and such factor must be unique by the fibre product in  $\mathbf{AffSch}$  shown above. So, it's indeed a fibre product in  $\mathbf{Sch}$ , not just in  $\mathbf{AffSch}$ .

□

To show it works for general schemes, it's similar to before: Restrict them to affine neighborhoods, construct fibre product neighborhood-wise, then glue all of them together. Where, the gluing of morphisms of schemes is done as follow: Given  $f_i : U_i \rightarrow Y$  and  $f_j : U_j \rightarrow Y$  two morphism of schemes (where  $U_i, U_j \subseteq X$  are open), then if  $f_i, f_j$  agrees on  $U_i \cap U_j$  as topological maps, they glue together

as  $f : U_i \cup U_j \rightarrow Y$  as topological maps. About the sheaves, for any open subset  $V \subseteq Y$ , its preimage  $f^{-1}(V) = f_i^{-1}(V) \cup f_j^{-1}(V)$ , which has the ring corresponding to the following fibre product diagram:

$$\begin{array}{ccc} \mathcal{O}(U_i \cup U_j) & \longrightarrow & \mathcal{O}(U_i) \\ \downarrow & & \downarrow \rho_{i,ij} \\ \mathcal{O}(U_j) & \xrightarrow{\rho_{j,ij}} & \mathcal{O}(U_i \cap U_j) \end{array}$$

Which, since  $f_i, f_j$  induces the ring homomorphisms  $f_i^* : \mathcal{O}_Y(V) \rightarrow \mathcal{O}(f_i^{-1}(V))$  and  $f_j^* : \mathcal{O}_Y(V) \rightarrow \mathcal{O}(f_j^{-1}(V))$ , then the fibre product property guarantees the following:

$$\begin{array}{ccccc} \mathcal{O}_Y(V) & & \xrightarrow{f_i^*} & & \mathcal{O}(U_i) \\ & \searrow \exists! f^* & & & \downarrow \rho_{i,ij} \\ & & \mathcal{O}(U_i \cup U_j) & \longrightarrow & \mathcal{O}(U_i) \\ & \searrow f_j^* & \downarrow & & \downarrow \rho_{i,ij} \\ & & \mathcal{O}(U_j) & \xrightarrow{\rho_{j,ij}} & \mathcal{O}(U_i \cap U_j) \end{array}$$

Where, the  $f^*$  commutes with other ring homomorphisms in a desirable way, which can be treated as the ring homomorphism associated to  $f : f_i^{-1}(V) \cup f_j^{-1}(V) \rightarrow V$  after the restriction. This generates a morphism of locally ringed space, because the fibre product of local homomorphisms is a local homomorphism (and same applies to the gluing of morphisms, since the maximal ideal of the fibre product is the intersection of the product maximal ideal in the product ring, with the subring identified as fibre product).

Now, let's talk about the most general case:

### *Theorem: Fibre Product of Schemes*

Given arbitrary scheme  $Z$ , and two schemes  $X, Y$  over  $Z$  (with morphisms  $f : X \rightarrow Z$  and  $g : Y \rightarrow Z$ ), then the fibre product in  $\text{Sch}$  exists.

**Proof:** Let's break it down to several steps:

#### **I. The Fibre Product of Open Subschemes:**

Let's temporarily assume fibre product  $X \times_Z Y$  exists, we wish to see how it decomposes to open subsets. So, assume we have this fibre product diagram in  $\text{Sch}$ :

$$\begin{array}{ccccc} & & f & & \\ & & \longleftarrow & & \\ Z & & & & X \\ \uparrow g & & & & \uparrow g' \\ Y & & & & X \times_Z Y \\ & & \longleftarrow f' & & \\ & & & & \end{array} \quad \begin{array}{c} \nearrow d_C \\ \searrow d_B \\ \text{---} \exists! h \end{array}$$

Given any open subscheme  $U \subseteq X$ , consider the open subscheme  $g'^{-1}(U) \subseteq X \times_Z Y$ , we claim that it's the fibre product  $U \times_Z Y$ :

Restrict the morphism  $f$  to  $f|_U : U \rightarrow Z$ , for any scheme  $W$  together with morphisms  $d_B : W \rightarrow Y$  and  $d_C : W \rightarrow X$  such that the image  $\text{im}(d_C) \subseteq U \subseteq X$ , and  $f|_U \circ d_C = g \circ d_B$ , then notice

without the restriction  $f \circ d_C = g \circ d_B$ , then using the fibre product property of  $X \times_Z Y$ , we get the unique morphism  $h : W \rightarrow X \times_Z Y$  satisfying the above diagram.

However, if consider  $W = d_C^{-1}(U) = h^{-1}(g'^{-1}(U))$ , this implies  $h(W) \subseteq g'^{-1}(U)$ , generating the morphism as desired:

$$\begin{array}{ccccc}
 Z & \xleftarrow{f|_U} & U & & \\
 \uparrow g & & \uparrow g'|_{g'^{-1}(U)} & & \searrow d_C \\
 Y & \xleftarrow{f'|_{g'^{-1}(U)}} & g'^{-1}(U) & & \\
 & \nwarrow f' & \nearrow f' & \nearrow \exists! h & \\
 & & & & W
 \end{array}$$

$d_B$

Finally, the reason  $h$  must be unique, is because any other  $h' : W \rightarrow g'^{-1}(U)$  with such property can also be embedded into the fibre product diagram of  $X \times_Z Y$ , and the universality there enforces it to be the same as  $h$ . Hence, we get that  $g'^{-1}(U) = U \times_Z Y$  as schemes.

As a last note, given extra  $V \subseteq Y$  open subschemes, repeating the argument using  $g'^{-1}(U) = U \times_Z Y$ , we get the fibre product  $U \times_Z V = f'^{-1}(V) \cap g'^{-1}(U)$  as open subschemes:

$$\begin{array}{ccccc}
 Z & \xleftarrow{f|_U} & U & & \\
 \uparrow g|_V & & \uparrow g'|_V & & \searrow d_C \\
 V & \xleftarrow{f'|_V} & f'^{-1}(V) \cap g'^{-1}(U) & & \\
 & \nwarrow f' & \nearrow f' & \nearrow \exists! h & \\
 & & & & W
 \end{array}$$

$d_B$

This asserts that fibre product is a „local property“, meaning restriction of fibre products onto open subsets of  $X, Y$  will necessarily generate a fibre product of the open subsets. Hence, with the gluing property of schemes, it suggests we can construct fibre products locally, then glue them together (once we guaranteed their compatibility).

## II. Gluing of Local Fibre Products:

This subsection will develop the compatibility locally. Given  $\{X_i\}$  an open cover of  $X$ , and suppose that for each  $i$ ,  $X_i \times_Z Y$  exists, we claim  $X \times_Z Y$  exists (more precisely, they can glue together to form  $X \times_Z Y$ ). We first assert the existence of this diagram:

$$\begin{array}{ccccc}
 Z & \xleftarrow{f|_{X_i}} & X_i & & \\
 \uparrow g & & \uparrow g'_i & & \searrow d_C \\
 Y & \xleftarrow{f'_i} & X_i \times_Z Y & & \\
 & \nwarrow f' & \nearrow f' & \nearrow \exists! h & \\
 & & & & W
 \end{array}$$

$d_B$

For each indices  $i, j$ , let  $X_{ij} := X_i \cap X_j$  open subscheme of  $X$ , we know  $g'^{-1}(X_{ij}) = X_{ij} \times_Z Y$  in  $X_i \times_Z Y$  by **I**, but similarly  $g'^{-1}(X_{ij}) = X_{ij} \times_Z Y$  in  $X_j \times_Z Y$ , while both are open. This

identifies an isomorphism of open subschemes  $g_i'^{-1}(X_{ij}) \cong g_j'^{-1}(X_{ij})$  between  $X_i \times_Z Y$  and  $X_j \times_Z Y$ , and hence the two glue together (including their morphisms, since these two restrictions must be compatible with the original morphisms  $f, g$ ).

Let  $P$  denotes the gluing of all  $X_i \times_Z Y$  via the above isomorphisms, with  $f' : P \rightarrow Y$  and  $g' : P \rightarrow X$  that is the gluing of all  $f'_i$  and  $g'_i$ 's respectively, we have the following commutative diagram:

$$\begin{array}{ccc} Z & \xleftarrow{f} & X \\ g \uparrow & & \uparrow g' \\ Y & \xleftarrow{f'} & P \end{array}$$

Such diagram commutes, because restricting to each  $X_i \times_Z Y$  (with  $f'_i, g'_i$  in the position of  $f', g'$  respectively) the diagram commutes, hence globally it also commutes.

Also, it satisfies the fibre product property: Given any scheme  $W$  with  $d_B : W \rightarrow Y$  and  $d_C : W \rightarrow X$  two morphisms of schemes which satisfies  $f \circ d_C = g \circ d_B$ , let  $W_i := d_C^{-1}(X_i)$  for each  $i$ , then it satisfies the following property:

$$\begin{array}{ccc} Z & \xleftarrow{f_i} & X_i \\ g \uparrow & & \uparrow g'_i \\ Y & \xleftarrow{f'_i} & X_i \times_Z Y \end{array} \quad \begin{array}{c} \nearrow d_{C,i} \\ \searrow \exists! h_i \\ \nearrow d_{B,i} \end{array} \quad \begin{array}{c} W_i \end{array}$$

Where, the  $h_i : W_i \rightarrow X_i \times_Z Y \hookrightarrow P$  is a morphism of schemes that agrees on each  $W_{ij} := W_i \cap W_j$  (since it's the preimage of  $X_{ij}$  under  $d_C$ ), so they should be compatible within  $X_{ij} \times_Z Y$  (which is the intersection of  $X_i \times_Z Y$  and  $X_j \times_Z Y$ ). When including each portion of the fibre square into the original space, it also satisfies the fibre product diagram.

Then, using the gluing property, there exists a unique  $h : W \rightarrow P$ , such that the following diagram commutes:

$$\begin{array}{ccc} Z & \xleftarrow{f} & X \\ g \uparrow & & \uparrow g' \\ Y & \xleftarrow{f'} & P \end{array} \quad \begin{array}{c} \nearrow d_C \\ \searrow \exists! h \\ \nearrow d_B \end{array} \quad \begin{array}{c} W \end{array}$$

The uniqueness of  $h$  is guaranteed, since its restriction onto each  $W_i$  must be  $h_i$  using the universal property of fibre product. Notice that each statement can also be applied on  $Y$  (when  $X$  is swapped with  $X_i$ ), hence as long as we have open covers  $\{X_i\}$  of  $X$  and  $\{Y_j\}$  of  $Y$ , such that each  $X_i \times_Z Y_j$  exists, then fibre product globally exists.

So, we've proven fibre product exists, once it locally exists.

### III. Case when $Z$ is Affine:

If  $Z$  is affine, choose an affine cover  $\{X_i\}$  of  $X$ ,  $\{Y_i\}$  of  $Y$ , then the restriction  $f_i : X_i \rightarrow Z$  and  $g_i : Y_i \rightarrow Z$  has fibre product  $X_i \times_Z Y_i$  (since they're all affine), and they agree on the intersection as before. So, they must glue to fibre product of  $X$  and  $Y$  by II.

**IV. General case for  $Z$ :**

Finally, If  $Z$  is arbitrary, let  $\{Z_i\}$  be an affine cover of  $Z$ , and take  $\{X_i := f^{-1}(Z_i)\}$  and  $\{Y_i := g^{-1}(Z_i)\}$  be the covering, then the restriction has  $f_i : X_i \rightarrow Z_i$  and  $g_i : Y_i \rightarrow Z_i$ , so **III** guarantees the existence of  $X_i \times_{Z_i} Y_i$ . Now, we claim that it's the same as  $X_i \times_Z Y$ : Given any scheme  $W$  together with morphisms  $d_B : W \rightarrow Y$  and  $d_C : W \rightarrow X_i$ , such that  $f_i \circ d_C = g \circ d_B$ . Then, notice that  $f_i \circ d_C(W) \subseteq f_i(X_i) \subseteq Z_i$ , showing that  $\text{im}(d_B) \subseteq g^{-1}(\text{im}(g \circ d_B)) \subseteq Y_i$ . Hence, restricting  $g$  to  $Y_i$  makes sense, we have a unique map  $h_i : W \rightarrow X_i \times_{Z_i} Y_i$ , such that the diagram commutes (when involving  $X_i, Y_i$ ).

But, since the map need no restriction, then it in fact satisfies the following commutative diagram:

$$\begin{array}{ccccc}
 & & Z & \xleftarrow{f_i} & X_i \\
 & & \uparrow g & & \uparrow f'_i \\
 & & Y & \xleftarrow{g'} & X_i \times_{Z_i} Y_i \\
 & & & \nwarrow g' & \nwarrow f'_i \\
 & & & & W
 \end{array}$$

$d_B$  (from  $W$  to  $Y$ ),  $d_C$  (from  $W$  to  $X_i$ ),  $\exists! h_i$  (from  $W$  to  $X_i \times_{Z_i} Y_i$ )

So,  $X_i \times_{Z_i} Y_i = X_i \times_Z Y$ , then by **II**, this guarantees the existence of  $X \times_Z Y$ , finishing the proof.

□

This again shows the general nature of algebraic geometry: restricting the situation to a smaller subset where one can guarantee nice behaviors (for instance, over affine schemes), and glue the local properties all together to a global one.

Also, the above has the following corollary:

**Corollary: Product of Schemes**

In the category **Sch**, product exists.

**Proof:** Since  $\text{Spec}(\mathbb{Z})$  is final in **AffSch**, together with Affinization, it's also final in **Sch**. Take the fibre product over  $\text{Spec}(\mathbb{Z})$ , we get the desired product. □

## 5 Fibre and Parameter Space

A general study of morphisms often includes its fibres. But, we do want to realize fibre not only as set theoretic definition, but also some geometric nature. With the existence of fibre product, we can then define such notion:

### Definition: Residue Field of a Point

Given a scheme  $Y$  and any point  $y \in Y$ , let  $\mathcal{O}_{Y,y}$  be its corresponding local ring. Then, the **residue field** of  $y$  is the residue field of  $\mathcal{O}_{Y,y}$ , denoted as  $k(y)$ .

Here, we claim there is a natural morphism  $\text{Spec}(k(y)) \rightarrow Y$ , given by  $* \mapsto y$ :

**Proof:** Choose an affine neighborhoods  $Y \supseteq U \ni y$ , then with  $U \cong \text{Spec}(A)$ , the local ring  $\mathcal{O}_{Y,y} \cong A_y$  (since here  $y$  represents a prime ideal). Hence, one has a natural composition  $A \rightarrow A_y = \mathcal{O}_{Y,y} \twoheadrightarrow k(y)$  (the first one is localization map, the second one is projection onto the residue field).

Then, this generates a natural morphism  $\text{Spec}(k(y)) \rightarrow \text{Spec}(A) = U$ , by  $* \mapsto y$  (as  $*$  is the zero ideal in  $k(y)$ , it pulls back to the maximal ideal of  $A_y$ , or  $y^e$  under localization; hence, it further pulls back to  $y \subseteq A$ ).

Which, with the inclusion morphism  $U \hookrightarrow Y$ , we naturally gets  $\text{Spec}(k(y)) \rightarrow Y$  by  $* \mapsto y$ .

As a side note, this is independent of the affine neighborhood, as the local ring is only dependent on the structure sheaf  $\mathcal{O}_Y$ .

□

Now, here's the formal definition of fibre:

### Definition: Fibre of a Morphism

Given a morphism of schemes  $f : X \rightarrow Y$ , and  $y \in Y$  be a point. Let  $k(y)$  be the residue field of  $y$ , and  $\text{Spec}(k(y)) \rightarrow Y$  be the natural morphism. Then, the **fibre** of  $f$  over the point  $y$  is the scheme:

$$X_y := X \times_Y \text{Spec}(k(y)) \quad (5.1)$$

which is given as the following fibre product diagram:

$$\begin{array}{ccc} X \times_Y \text{Spec}(k(y)) & \xrightarrow{f'} & \text{Spec}(k(y)) \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

Intuitively, as topological spaces one must have the fibre product here to be  $f^{-1}(y)$  (since one of the space is one point, hence it shouldn't change the product structure). But, here there are extra algebraic structures one needs to worry about. Let's prove this claim:

**Proof:** When constructing the morphism  $\text{Spec}(k(y)) \rightarrow Y$ , we've already used the existence of affine neighborhood of  $y \in Y$ , so here WLOG we'll say  $Y = \text{Spec}(A)$  is affine (by restricting  $X$  to suitable preimage, and  $Y$  to the affine neighborhood).

Now, let  $\{U_i\}$  be an affine cover of  $X$ , each with a morphism  $f_i : U_i \rightarrow Y$ . We'll first prove that the fibre product  $U_i \times_Y \text{Spec}(k(y))$  is homeomorphic to  $f^{-1}(y) \cap U_i$ .

Since  $U_i, Y$  are now affine, say  $U_i = \text{Spec}(B_i)$ , one realizes  $U_i \times_Y \text{Spec}(k(y)) = \text{Spec}(B_i \otimes_A k(y))$ , where the morphisms are as follow (left in Sch, right in CRing):

$$\begin{array}{ccc} \text{Spec}(B_i \otimes_A k(y)) & \longrightarrow & \text{Spec}(k(y)) \\ \downarrow & & \downarrow \\ \text{Spec}(B_i) & \longrightarrow & \text{Spec}(A) \end{array} \quad \begin{array}{ccc} B_i \otimes_A k(y) & \longleftarrow & k(y) \\ \uparrow & & \uparrow \\ B_i & \longleftarrow & A \end{array}$$

Which, realizing  $y \subseteq A$  as a prime ideal the problem translates to: Is there a homeomorphism between  $\{\text{prime ideals } P \subseteq B_i \text{ that contracts to } y\}$  and  $\{\text{prime ideals } Q \subseteq B_i \otimes_A k(y)\}$ .

Recall that  $k(y) := A_y / (yA_y)$  (where  $A_y$  is the localization at prime ideal  $y$ , and  $yA_y$  is the associated prime ideal). Then, given the exact sequence  $0 \rightarrow yA_y \rightarrow A_y \rightarrow k(y) \rightarrow 0$ , tensoring with  $B_i$  realizes the following right exact sequence:

$$B_i \otimes_A (yA_y) \rightarrow B_i \otimes_A A_y \cong (B_i)_y \rightarrow B_i \otimes_A k(y) \rightarrow 0 \quad (5.2)$$

Where  $(B_i)_y$  is viewed as localization of  $B_i$  at  $y$  as  $A$ -module (which itself has an algebra structure). Then, the image of  $B_i \otimes_A (yA_y)$  in  $(B_i)_y$  is precisely  $y(B_i)_y$  (the extension of  $y$  in  $(B_i)_y$ ), so one has  $B_i \otimes_A k(y) \cong (B_i)_y / y(B_i)_y$ .

Hence, the prime ideals of  $B_i \otimes_A k(y)$  is precisely all prime ideals in  $(B_i)_y$  that contains  $y(B_i)_y$ , and these must contract to  $y$  (as it contains image of  $y$ , while the contraction can't be anything larger than  $y$ , or else the contraction then extension will be the unit ideal).

Finally, these prime ideals in  $B_i / yB_i$

□